

SEMENIVKA, Ukraine

WMO: 330490

Lat: 52.1908N

Lon: 32.5739E

Elev: 528

StdP: 14.42

Time Zone: 2.00 (EUE)

Period: 86-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.9	1.2	-9.9	3.3	-3.4	-4.3	4.5	1.9	27.0	28.6	23.5	28.1	5.9	20	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	86.0	66.9	82.5	66.0	79.4	64.8	69.5	80.4	67.8	78.6	66.3	77.3	6.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.1	98.2	73.6	63.9	90.6	71.9	62.3	85.6	70.5	33.8	81.1	32.4	78.4	31.2	77.6	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	16.3	14.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.6	22.2	25.4	33.3	46.1	57.5	62.8	65.3	64.2	55.1	42.8	33.9	25.9
	DBStd	17.64	11.06	10.79	9.60	7.87	7.32	5.76	5.02	6.74	7.30	7.94	9.63	9.47
	HDD50	3761	862	689	520	168	15	0	0	0	28	248	485	746
	HDD65	7650	1327	1109	983	567	252	110	59	102	309	688	933	1211
	CDD50	1806	0	0	2	50	249	383	475	440	180	25	2	0
	CDD65	222	0	0	0	0	21	44	69	77	11	0	0	0
	CDH74	2408	0	0	0	6	224	363	672	1024	119	0	0	0
	CDH80	636	0	0	0	0	32	67	142	375	20	0	0	0
Wind	WSAvg	6.2	8.0	6.9	7.6	6.7	5.7	5.4	5.3	4.7	5.1	5.6	6.1	7.7
Precipitation	PrecAvg	30.7	3.1	2.5	2.4	1.8	2.6	3.0	3.5	2.1	2.3	2.2	2.4	2.9
	PrecMax	42.0	6.6	5.0	5.7	4.0	4.9	6.3	6.0	4.6	4.7	4.6	4.1	7.3
	PrecMin	23.3	1.1	0.9	0.3	0.1	1.3	1.5	1.1	0.1	0.4	0.2	0.2	0.7
	PrecStd	3.8	1.2	1.0	1.3	0.9	0.9	1.1	1.4	1.1	1.2	1.3	1.1	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.7	48.7	62.2	72.5	82.8	86.1	86.6	92.6	82.7	69.7	56.9	47.0
		MCWB	39.5	43.0	49.6	57.0	63.5	68.6	67.4	67.3	66.5	57.6	53.9	43.1
	2%	DB	39.1	41.6	55.1	68.9	79.8	81.5	83.6	88.2	77.8	62.7	51.3	44.0
		MCWB	37.1	38.5	46.4	54.1	63.9	66.7	66.8	66.7	62.4	56.0	49.0	41.7
	5%	DB	37.0	38.5	49.8	64.5	75.8	78.1	80.6	84.2	72.9	58.3	47.8	40.0
		MCWB	35.1	36.2	42.6	51.5	62.3	64.9	66.0	66.6	60.1	53.4	45.7	38.4
	10%	DB	35.0	36.5	45.5	59.8	71.3	74.6	77.3	79.6	68.1	54.8	44.4	36.0
		MCWB	33.5	34.7	40.3	49.8	59.4	62.2	65.0	63.8	58.4	51.1	42.5	34.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.5	43.8	50.1	59.0	69.3	71.4	72.7	70.4	67.1	61.2	54.0	44.8
		MCDB	41.3	47.1	59.9	73.7	77.7	80.2	80.4	85.0	80.1	65.9	56.5	45.7
	2%	WB	37.7	39.1	47.2	55.5	65.6	68.7	69.5	68.2	64.0	56.7	49.6	41.5
		MCDB	38.7	41.4	54.2	64.9	76.1	78.1	79.1	82.8	75.3	61.3	50.8	43.6
	5%	WB	35.4	36.8	44.0	53.4	63.1	66.8	67.7	66.8	61.4	54.0	46.1	38.2
		MCDB	36.8	38.2	49.0	62.1	73.0	75.1	76.7	81.1	69.5	57.1	47.7	40.2
	10%	WB	33.3	34.9	41.0	51.3	61.0	64.5	66.0	65.3	59.0	51.8	42.8	34.7
		MCDB	34.5	36.2	45.0	58.7	69.6	72.6	75.4	78.0	67.5	54.8	44.4	35.6
Mean Daily Temperature Range	5% DB	MDBR	8.4	9.9	12.9	16.3	19.5	17.7	19.2	22.2	17.1	14.0	8.2	7.8
		MCDBR	5.6	9.9	18.7	24.4	24.1	22.0	24.5	29.6	25.8	20.5	9.9	10.4
		MCWBR	5.4	7.5	11.4	12.2	11.3	9.7	10.5	11.0	13.3	12.6	7.7	9.3
	5% WB	MCDBR	5.7	9.0	17.4	21.2	20.7	19.1	20.6	26.8	23.2	16.4	8.9	10.3
		MCWBR	5.7	7.5	11.2	11.7	10.6	9.6	10.1	11.0	13.0	12.1	7.5	9.2
Clear-Sky Solar Irradiance	taub	0.301	0.334	0.364	0.430	0.402	0.389	0.423	0.427	0.390	0.359	0.338	0.305	
	taud	2.140	2.075	2.137	2.121	2.284	2.347	2.270	2.241	2.304	2.355	2.259	2.181	
	Ebn at Noon	211	238	258	254	270	274	262	253	248	232	198	187	
	Edh at Noon	23	33	38	44	39	37	40	39	33	25	21	19	
All-Sky Solar Radiation	RadAvg	257	539	975	1315	1773	1872	1781	1548	1031	537	215	157	
	RadStd	51	86	148	119	134	160	128	140	119	94	39	37	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	+1.17	N/A	N/A	+1.67	+1.01	+0.95	-248	-378	+180	+67

Nomenclature: See separate page